Module-2

Web Analytics Process

The primary objective of carrying out Web Analytics is to optimize the website in order to provide better user experience. It provides a data-driven report to measure visitors‟ flow throughout the website.

Web analytics is the measurement, collection, analysis, and reporting of web data to understand and optimize web usage. It helps businesses and website owners make data-driven decisions to improve user experience, marketing effectiveness, and website performance.

A typical web analytics model involves:

- **Data Collection**: Gathering raw data from user interactions (page views, clicks, conversions).
- **Data Processing**: Organizing and structuring the raw data for analysis.
- **Data Analysis**: Applying metrics, segmentation, and visualization to uncover patterns and insights.
- **Action**: Making decisions and optimizing strategies based on insights.

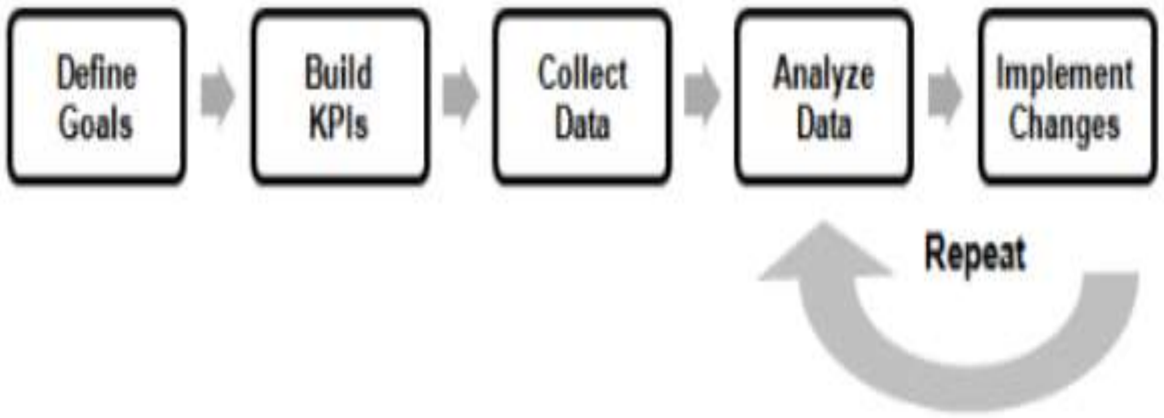
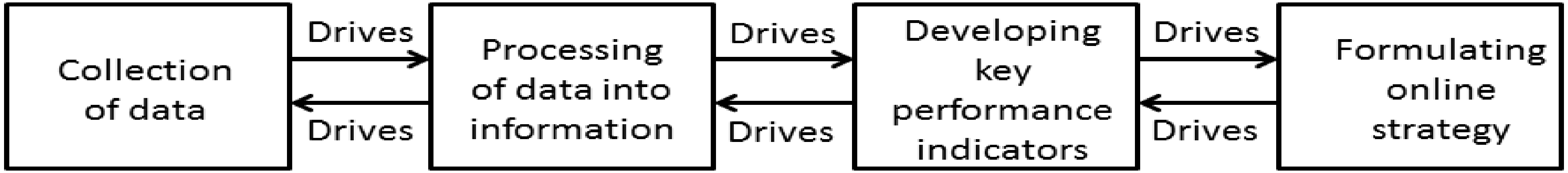


Figure 1: The Web Analytics Process. Source: [2]

Web Analytics process starts with defining goals for the Website, then there should be concrete KPIs (Key Performance Indicators) that will help understand the different metrics used in the Web Analytics process.

After step 1 and 2 has been clearly defined, it is time to collect the data, and there are several way to collect data in the context of Web Analytics . After collecting the data comes analyzing the data part, and finally implementing the changes based on the analysis results which corresponds to the goals and KPIs

Basic Steps of Web Analytics Process



Collection of data	Processing of data into information	Developing key performance indicators	Formulating online strategy
Typically, counts. Basically, data collection	Typically, ratios. Data becomes metrics.	Counts and ratios infused with business strategy.	Online goals, objectives, or standards for organization.
Examples: • Time stamp • Referral URL • Query terms	Examples: • Time on page • Bounce rate • Unique visitors	Examples: • Conversion rate • Average order value • Task completion rate	Examples: • Save money • Make money • Marketshare

- **Data collection**:

This is the foundation of web analytics, where raw data is gathered from user interactions with the website. For example:

  - Time stamps: When users visit or interact with the site
  - Referral URLs: Where users came from (e.g., search engines, social media)
  - Query terms: What users searched for to find the site
  - Page views: Which pages users visited
  - User-agent strings: Information about users' devices and browsers

- **Turning data into insights:**

Data is turned into actionable insights using metrics that describe user behavior. Examples of metrics include:

  - Bounce rate: Percentage of single-page visits
  - Unique visitors: Number of individual users visiting the site
  - Page views per session: How many pages users view on average

- **Formulating strategy:**

Businesses set objectives and create action plans to achieve desired results.

Examples of strategic goals include:

  - Increasing sales by a certain percentage
  - Gaining a larger market share
  - Improving customer retention rates
  - Enhancing user engagement on the website

- **Developing Key Performance Indicators (KPIs)**

Metrics are combined with business objectives to create KPIs that measure success. It includes:

  - Average order value: Typical amount spent per transaction
  - Task completion rate: How often users successfully complete a specific task
  - Customer lifetime value: Predicted net profit from the entire future relationship with a customer

Web analytics tools

- **Google Analytics.** Google Analytics is a web analytics platform that monitors website traffic, behaviors and conversions. The platform tracks page views, unique visitors, bounce rates, referral Uniform Resource Locators, average time on-site, page abandonment, new vs. returning visitors and demographic data.

- **Optimizely.** Optimizely is a customer experience and A/B testing platform that helps businesses test and optimize their online experiences and marketing efforts, including conversion rate optimization.

Kissmetrics. Kissmetrics is a customer analytics platform that gathers website data and presents it in an easy-to-read format. The platform also serves as a customer intelligence tool, as it enables businesses to dive deeper into customer behavior and use this information to enhance their website and marketing campaigns.

- **Crazy Egg.** Crazy Egg is a tool that tracks where customers click on a page. This information can help organizations understand how visitors interact with content and why they leave the site. The tool tracks visitors, heatmaps and user session recordings.

Common issues with web analytics

- Keeping track of too many metrics
- Data is not always accurate
- Data privacy is at risk
- Data does n‟t tell the whole story

Showcasing the Work

Presenting web analytics involves:

- **Dashboards and reports highlighting KPIs (Key Performance Indicators).**
- Visualizations like charts, graphs, and heatmaps to communicate findings clearly.
- Case studies demonstrating how analytics led to improvements or ROI.
- Storytelling to connect data insights with business goals.

Context Matters with Data

Data never exists in a vacuum. Understanding the context is critical:

- What are the business objectives?
- Who is the target audience?
- What external factors (seasonality, campaigns) affect the data?
- Technical setups and tracking limitations.
  Context helps interpret data correctly and avoid misleading conclusions.

Contradicting the Data

- Sometimes data can appear contradictory or confusing:
- Different tools may report varying numbers due to tracking methods.
- User behaviour might not align with business expectations.
- Outliers or anomalies can skew interpretation.
  To resolve contradictions, validate data sources, cross-reference metrics, and consider qualitative insights (e.g., user feedback).

Page Tagging (cookies) vs. Log Analysis

- Web analytics helps us understand how users interact with a website. There are different ways you can collect data about your website visitors. One way is to analyze the log files your web server creates. The other way is often called page tagging. This involves placing javascript tags in your website code. This javascript sends the tracking data back to some analytics tool.

Two primary data collection methods are:

Log File Analysis vs. Page Tagging

Page Tagging

- *Page tagging* collect data via the visitor's web browser and send information to remote data-collection servers.
- The analytics customer views reports from the remote server.This information is usually captured by JavaScript code (known as *tags* or *beacons*) placed on each page of your site.
- Some vendors also add multiple custom tags to collect additional data. This technique is known as *client-side data collection* and is used mostly by outsourced, Software as a Service (SaaS) vendor solutions.

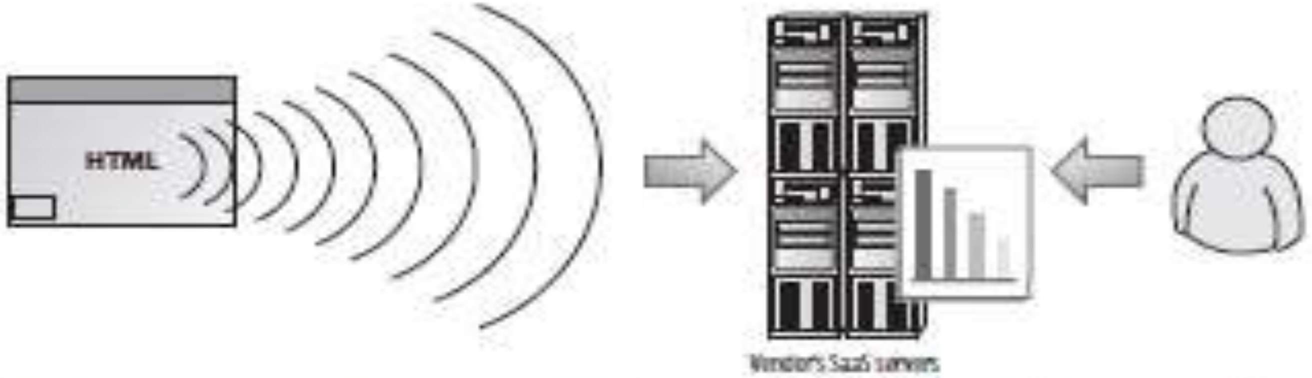


Figure 2.1 Schematic page tag methodology: Page tags broadcast information to remote data-collection servers, thus enabling the analytics customer to view reports.

Log files

- *Log files* refer to data collected by your web server independently of a visitor's browser: The web server logs its activity to a text file that is usually local.
- The analytics customer views reports from the local server, as shown in Figure. This technique, known as *server-side data collection*, captures all requests made to your web server, including pages, images, and PDFs, and is most frequently used by stand-alone licensed software vendors.

Figure 2.2 Schematic logfile methodology: The web server logs its activity to a text file locally, thereby enabling the analytics customer to view the reports on the local server.

- Dashboards and reports highlighting KPIs (Key Performance Indicators).

Dashboards

- A **dashboard** is a visual tool that displays key data and metrics (KPIs) in real time to help monitor performance and make quick decisions.
- To provide real-time, actionable insights for quick decision-making.
- Dashboard is an interface showing graphical status of the trends of your business key performance indicators. This helps you to take instantaneous and intelligent decisions. It gives you a visual display of important data that can be encapsulated in a single space to let you monitor in a glance.

- **Common Components**:
    - **Charts & Graphs**: Bar charts, line graphs, pie charts, etc.
    - **Gauges & Indicators**: Show performance status (e.g., green = good, red = bad).
    - **Filters/Interactivity**: Allow users to customize views by time, region, product, etc.

Web Analytics Dashboards

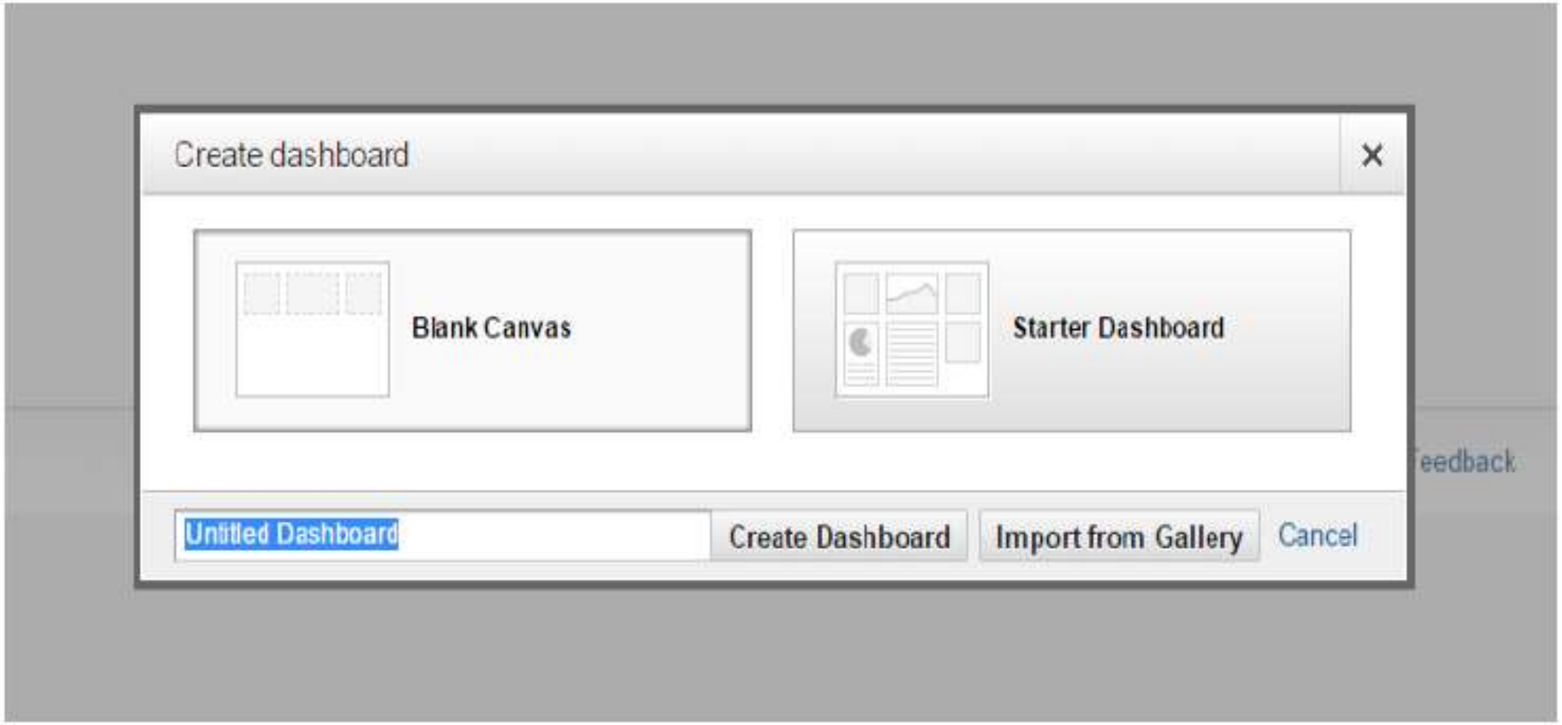
- Web analytics dashboard is like a control panel that displays the most important information and data related to your website's performance. The best way to understand what this dashboard looks like is to picture the control panel of your vehicle.
- A web analytics dashboard presents a snapshot of the lengthy analysis conducted using complicated tools. It is like a condensed and summarized report that lets a business representative assess how well the website has been doing.

Online Visitors
45
67
46
Today's Overview
1,647
35.7%
3,790
14m
14,696
3.8
Today Vs Trend
3,792
New Vs Returning
142,737
Live Map
45
Live Referrers
google.com.au 2
google.co.in 1
1
1
Live Events
Web View 38
Pageview 7
Login 1
app-log 1
App Warning 1
Article View 1
Live Pages
2
/ 1
1
1
1
1

Dashboard Implementation

- In Google analytics, you can create dashboards according to your requirements. Dashboards are used for finding data. With the help of dashboards, you can quickly analyze the data. In dashboard, you have to create widgets as per your requirements.
- A *dashboard* builder is a website analytics tool to help you build a dashboard that will showcase data and provide a realtime view of your web analytics metrics.

The following image shows how to create a dashboard:



Why use web analytics dashboard?

- Web analytics dashboard is a built-in feature in analytical tools and can be accessed by logging in to the software used for assessment. Having such a dashboard to refer to is definitely a big plus because the numbers and results that matter the most to your need can be easily kept at hand, without going through lengthy analytical reports.

It lets users:

- Set goals that are to be achieved with the website.
- Track progress on a daily, weekly and monthly basis
- Identify danger signals that can be spotted immediately once the right figures are highlighted.
- Decide on corrective action by isolating the problem areas

Types of Web Analytics Dashboards

You can create dashboards according to your requirements.

Following are the main types of dashboards:

- SEO dashboard
- Content dashboard
- Website performance dashboard
- Real time overview dashboard
- Ecommerce dashboard
- Social Media dashboard
- PPC dashboard

Metrics for Every Dashboard

- **Search Engine Optimization (SEO)** – Organic traffic, Website total traffic, Keyword used in Organic, Top landing pages, etc.
- **Content** – In content dashboard, you have to monitor traffic for blog section, Conversion by blog post, and Top landing page by exit.
- **Website Performance Dashboard** – Avg. page load time, Mobile page load time, Page load time by browser, and Website server response time.
- **Real Time Overview Dashboard** – In real time overview, you can set a widget for real time traffic, Real time traffic source, and real time traffic landing pages.
- **Ecommerce Dashboard** – In ecommerce total traffic, Landing by products, and Total sale by products.
- **Social Media Dashboard** – In social media traffic by social media channel, Sale by social media, most socially shared content.
- **PPC dashboard** – In pay per click (PPC) dashboard, you need to include clicks, impressions, CTR(click through rate), converted clicks, etc.

Report

A report is a structured document that presents detailed data and analysis, often over a specific time period, to evaluate performance or trends.

Types of Reports in Web Analytics

- Traffic Reports
- Acquisition Reports
- Behavior Reports
- Conversion Reports
- Audience Reports
- Real-Time Reports
- Event Tracking Reports
- Landing Page Reports

- **Traffic Reports**

Traffic reports provide an overview of the number of visitors to a website and how those visitors engage with it. These reports include metrics such as pageviews, sessions, users, average session duration, and bounce rate. By analyzing traffic reports, organizations can identify peaks or drops in website activity, assess the impact of marketing campaigns, and understand general user engagement trends over time.

- **Acquisition Reports**

Acquisition reports focus on how visitors arrive at a website. They break down traffic sources into categories such as organic search, direct visits, referral traffic, social media, email campaigns, and paid advertisements. These reports are essential for evaluating the effectiveness of marketing strategies and optimizing channels that bring the highest quality traffic.

- **Behavior Reports**

Behavior reports delve into how users interact with the website once they arrive. They highlight the most visited pages, time spent on each page, bounce rates, exit pages, and user flow. These insights help website owners understand which content performs well, which pages cause users to leave, and how the site structure influences navigation.

- **Conversion Reports**

Conversion reports measure how well a website meets its business objectives, such as lead generation, form submissions, downloads, or purchases. These reports track goal completions and conversion rates, providing crucial data on the performance of landing pages, marketing funnels, and calls to action. For e-commerce websites, they also include data on transactions, revenue, and average order value.

- **Audience Reports**

Audience reports give demographic and technical information about website users. They include data on users' age, gender, location, interests, devices used (desktop, mobile, tablet), browser types, and whether the user is new or returning.

- **Real-Time Reports**

Real-time reports show live activity on a website, including the number of active users, their geographic locations, current pageviews, and traffic sources. These reports are useful during marketing campaigns, product launches, or live events, as they allow immediate monitoring of visitor behavior and campaign performance.

- **Event Tracking Reports**

Event tracking reports monitor specific user interactions that are not captured by default, such as clicks on buttons, downloads, video plays, and form submissions. These interactions, known as "events," provide deeper insights into how users engage with interactive elements on a site, allowing for more detailed behavioral analysis and UX optimization.

- **Landing Page Reports**

Landing page reports assess the performance of entry points to the website. They show which landing pages attract the most visitors and how well those pages convert users. These reports help identify high-performing entry pages and those that may need improvements in content, design, or loading speed to increase conversions.

What is Google Analytics?

- A free web analytics tool by Google
- Measures and reports website/app traffic
- Provides actionable data on user behavior
- Google Analytics (GA) as a **powerful tool for digital marketers, developers, and business teams**. It collects user interaction data such as page views, time spent, user flow, and device types. Emphasize its **real-time capabilities** and that it’s **used worldwide** to make better website and marketing decisions.

Why Use Google Analytics?

- Understand how users interact with your website
- Identify sources of traffic
- Optimize user experience (UX)
- Improve campaign performance
- Increase conversions and revenue

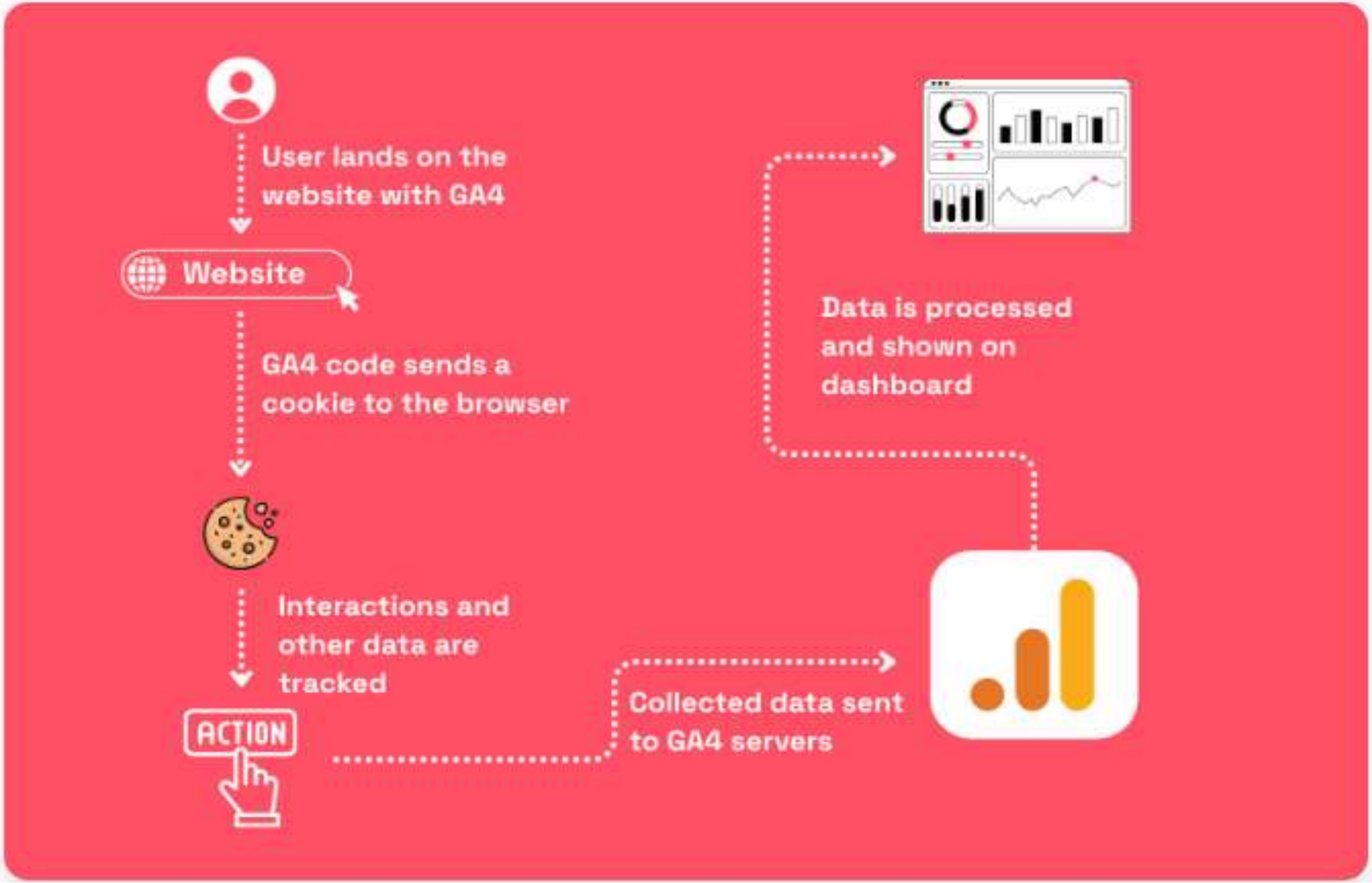
Key Features

- Real-time tracking
- Audience insights (demographics, devices, locations)
- Acquisition analysis (traffic sources)
- Behavior tracking (pageviews, time on site)
- Conversion tracking (goals, eCommerce, events)

How Google Analytics collects data

- Google Analytics collects data through short lines of Javascript or HTML code called **tags**.
- These tags are placed on a web page and collect **data points** when triggered by a specific event- for instance, when a visitor views a page or clicks on a link.
- Google Analytics processes these data points to provide users with **metrics**. For instance, page views are used to count visits, and these data are then shown in a dashboard to tell you how many visitors came to your website this week.
- Google Analytics organizes these metrics into **reports** to give the user an overview of the performance of their website.

How Google Analytics collects data



Google Analytics datapoints collection

Google Analytics tracks various hits (or datapoints), including pageviews, events, and transactions.

- **Pageviews** are the most basic type of hits. They represent a user visiting a page on your website. Every time a visitor loads a page on your website, a pageview hit is recorded in Google Analytics.
- **Events** are interactions with specific elements on your website such as clicking on a button or playing a video. You can use events to track user behavior and get more detailed insights into how visitors interact with your website.
- **Transactions** are completed purchases or other conversions on your website. They represent the result of a user's journey on your website, and they are often used to measure the success of your website's performance. Transactions are mainly used by e-commerce websites and are sometimes referred to as e-commerce tracking.

METRICS DOES GOOGLE ANALYTICS COLLECT

Google Analytics metrics collection

Using the datapoints Google Analytics organizes a large range of metrics that are shown in your dashboard. Here are some of the most important ones:

- **User demographics**: Google Analytics collects data about the age, gender, and interests of your users, as well as their location and language. This metric is processed using data from third-party sources as well, such as Google's advertising network. The practice is **highly privacy-invasive** as you collect personal information on your website visitors. If you don’t need this, you can also look at other analytics tools that are more privacy-friendly.
- **Unique visitors**: Google Analytics can tell whether a page view comes from a first-time or returning visitor. This metric can help you assess whether your audience is growing and which pages are the most successful in attracting new visitors to your website.

Google Analytics metrics collection

- **Traffic sources**: data is collected on the sources of your website traffic, including data about the search engines, websites, and social networks that are referring traffic to your website. This data helps you understand how people find your website.
- **Traffic channels**: this metric is similar to traffic sources but not quite the same. Traffic channel is based on classifying traffic sources into different groups, such as organic search, direct traffic, referral traffic, and paid search.
- **Conversions and goals**: the customer can set up specific goals in Google Analytics. Goals and conversion data are collected, including the number of times a goal is completed, the value of each goal, and the conversion rate for each goal.

- **Device and browser information**: Type of devices, operating systems, and web browsers. This data can help optimize the user experience for different devices and browsers.
- **User engagement**: GA collects data on how engaged your website visitors are, based on how many pages they view, the time they spend on each page, and the bounce rate (the percentage of users who leave after viewing only one page).
- **Campaigns and promotions**: data about the campaigns that you are running for your website, including data about the keywords, ads, and landing pages that are associated with each campaign.

Google Analytics and cookie consent

- Google Analytics is built around cookies. Cookies are small files that Google Analytics reads and writes on a user's browser. Google Analytics cookies include **unique identifiers**. This identifier allows Google Analytics to recognize an individual user by simply reading their cookies.
- Cookies allow Google Analytics to refer data points to the same visitor. Some key metrics simply cannot be collected without cookies! This is why **Google Analytics 4 still uses first-party cookies** despite being advertised by Google as a cookieless tool.
- For instance, knowing what pages attract the newest visitors to your website can be useful. But Google Analytics cannot count unique visitors without using cookies. It will still register each page view, but because it cannot link them with a unique ID, it will have no idea whether each page view came from a unique or a returning visitor.

Interacting with Data in Google Analytics

- In the digital age, data is one of the most valuable assets for any business operating online. However, the true value of data is not in its mere collection, but in how it is analyzed, interpreted, and applied to decision-making. This is where the concept of **interacting with data** in **Google Analytics** becomes essential.

- Particularly with the advent of **Google Analytics 4 (GA4)**, interaction with data has evolved from static reporting to dynamic exploration, enabling businesses to better understand their users, measure performance, and uncover actionable insights.

Understanding the Importance of Data Interaction

- Google Analytics is a powerful tool that collects vast amounts of information about user behavior, acquisition sources, engagement, conversions, and more. But data alone does not answer business questions.
- By **interacting with data**, users can explore patterns, segment audiences, apply filters, compare metrics, and customize views to find meaningful insights that support business goals.
- For example, a spike in traffic might seem like a success at first glance. However, by digging deeper into the data — such as identifying traffic sources, analyzing bounce rates, or comparing engagement across devices — an analyst may find that most of this traffic is irrelevant or unqualified. Without interacting with the data, such distinction would be easily overlooked.

Core Methods of Interacting with Data in Google Analytics

Using the Report Interface

- GA4 provides a range of standard reports categorized by user lifecycle stages: acquisition, engagement, monetization, and retention. These reports are designed to offer high-level overviews as well as detailed breakdowns.
- Users can interact with these reports by adjusting date ranges, applying comparisons, sorting by metrics, and clicking on dimensions (e.g., country, device type) to drill down into more specific insights.
- This allows for a flexible and user-friendly way to explore performance metrics across various touchpoints.

Applying Filters and Segments

- One of the most powerful ways to interact with data is through **filters** and **segments**.
- Filters allow users to narrow down data views based on criteria such as traffic source, device category, or geographic location.
- Segments, on the other hand, enable comparisons between different subsets of users — such as new vs. returning visitors or mobile vs. desktop users. These techniques make it possible to isolate specific behaviors and understand how different user groups engage with content, navigate the site, and convert.
- For instance, a company may want to see how users from a Facebook campaign compare to those from Google Ads in terms of session duration or conversions. By creating segments for each traffic source, the business can analyze and optimize marketing performance based on data-driven insights.

Customizing Reports

- GA4 allows for extensive report customization. Users can modify existing reports or create entirely new ones tailored to their specific objectives.
- This includes choosing different metrics and dimensions, changing visualization types (such as tables, bar graphs, or line charts), and saving views for frequent analysis.
- Custom reports ensure that stakeholders are focusing on the metrics that matter most to them, avoiding unnecessary data clutter and improving efficiency in decision-making.

Explorations in GA4

The **Explorations** section in GA4 provides advanced analysis capabilities, offering tools such as **free-form tables**, **funnel analyses**, **path exploration**, **segment overlap**, and **cohort analysis**. These tools are designed for deep dives into user behavior and conversion paths.

Funnel analysis shows the steps users take towards conversion, helping identify where drop-offs occur.

Path exploration visually maps user journeys through the website, revealing navigation patterns.

Segment overlap highlights the intersections between user groups with shared traits or behaviors.

By manipulating variables and applying filters in real time, analysts can test hypotheses, discover trends, and make data-backed optimizations.

Events and Conversions

GA4 has transitioned from a session-based model to an event-based one. This means everything a user does — from viewing a page to clicking a button or watching a video — can be tracked as an event. Users can define custom events, set up event parameters, and mark certain events as **conversions** to track key business goals.

By interacting with event data, businesses can answer important questions:

What actions do users take before converting?

Which interactions indicate high engagement?

What features are being ignored?

This interactive tracking allows for precise measurement of user behavior and the effectiveness of site features or marketing campaigns.

Reference

Advanced Web Metrics with Google Analytics™ Second Edition Brian Clifton